SE LAB 7

Name: Krishitha Kankanala

SRN: PES2UG24CS234

Class: 5D

Original Output

```
PS C:\Users\Krishitha\L6_L7_44_scholarship_awards\LAB7> python scholarship_awards.py
==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:82
Category and Marks: ('Merit', 82.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 72,85,90
Average Marks: 82

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 5000|500
Payable Award: 5400.0
PS C:\Users\Krishitha\L6_L7_44_scholarship_awards\LAB7>
```

Fuzz Testing Failure

```
=========================== short test summary info ============================
FAILED test_scholarship_awards.py::test_parse_applicant_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_scholarship_awards.py::test_marks_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_scholarship_awards.py::test_payable_award_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 7.03s ===============================
```

Fixed Application

```
PS C:\Users\Krishitha\L6_L7_44_scholarship_awards\LAB7> python scholarship_awards.py
==== Running parse_applicant ====
Enter category:marks (e.g. merit:82): merit:82
Category and Marks: ('Merit', 82.0)

==== Running marks_average ====
Enter marks as CSV (e.g. 72,85,90): 72,85,90
Average Marks: 82

==== Running payable_award ====
Enter award|deduction (e.g. 5000|500): 5000|500
Payable Award: 5400.0
```

Fuzz Testing Passed

```
PS C:\Users\Krishitha\L6_L7_44_scholarship_awards\LAB7> python -m pytest test_scholarship_awards.py -v
============================= test session starts =============================
platform win32 -- Python 3.12.7, pytest-8.3.5, pluggy-1.6.0 -- C:\Users\Krishitha\AppData\Local\Programs\Python\Python312\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Krishitha\L6_L7_44_scholarship_awards\LAB7
plugins: anyio-4.16.0, hypothesis-6.170.15, langsmith-0.4.20
collected 3 items

test_scholarship_awards.py::test_parse_applicant_no_crash PASSED         [ 33%]
test_scholarship_awards.py::test_marks_average_no_crash PASSED           [ 66%]
test_scholarship_awards.py::test_payable_award_no_crash PASSED           [100%]
```